

Recall from last class:

- **Theorem:** Let $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ be bases of a vector space V . Then there exists a unique $n \times n$ matrix P such that $[x]_{\mathcal{C}} = P[x]_{\mathcal{B}}$.
- We call P the **change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** . In general

$$P = [[b_1]_{\mathcal{C}} \ \dots \ [b_n]_{\mathcal{C}}].$$

Also P^{-1} is the change of coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

1. Let V be a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and W be a vector space with basis $\mathcal{C} = \{c_1, \dots, c_m\}$. Given a linear transformation $T : V \rightarrow W$, we naturally seek an associated matrix to represent T using the bases; namely we want an $m \times n$ matrix A such that $[T(x)]_{\mathcal{C}} = A[x]_{\mathcal{B}}$ for all $x \in V$. How can we find the matrix A ?

We call A the **matrix of T relative to bases $\mathcal{B}$ and $\mathcal{C}$** . In general $A = [[T(b_1)]_{\mathcal{C}} \ \dots \ [T(b_n)]_{\mathcal{C}}]$.

2. Let $\mathcal{B} = \{b_1, b_2, b_3\}$ and $\mathcal{D} = \{d_1, d_2\}$ be bases for V and W respectively. Let $T : V \rightarrow W$ be the linear transformation such that $T(b_1) = 3d_1 - 5d_2$, $T(b_2) = -d_1 + 6d_2$, and $T(b_3) = 4d_2$. Find the matrix of T relative to $\mathcal{B}$ and $\mathcal{D}$.

3. Let $T : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ by $T(p) = \frac{dp}{dt}$. Find the matrix of T relative to the standard basis.

4. Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x) = Ax$ and $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$.

(a) Find the matrix of T relative to $\mathcal{B}$.

(b) Find A^{2019} .

Diagonal Representation Theorem: Suppose $A = PDP^{-1}$ where D is a diagonal matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed by the columns of P , then D is the matrix of T relative to $\mathcal{B}$.

5. Prove the theorem.

Definition 1. If A and B are $n \times n$ matrices, then A is **similar** to B if there exists an invertible matrix P such that $A = PBP^{-1}$.

Definition 2. If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v , called an **eigenvector** corresponding to λ , such that

$$Av = \lambda v.$$

Eigenvalues and Eigenvectors – Answers / Solutions

1. Recall the matrix M of T relative to bases $\mathcal{B}$ and $\mathcal{D}$ is given by $M = [[T(b_1)]_{\mathcal{D}} \ [T(b_2)]_{\mathcal{D}} \ [T(b_3)]_{\mathcal{D}}]$. Since $T(b_1) = 3d_1 - 5d_2$, we get $[T(b_1)]_{\mathcal{D}} = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$. Since $T(b_2) = -d_1 + 6d_2$, we get $[T(b_2)]_{\mathcal{D}} = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$. Since $T(b_3) = 4d_2$, we get $[T(b_3)]_{\mathcal{D}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Thus the matrix of T relative to bases $\mathcal{B}$ and $\mathcal{D}$ is given by $\begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix}$.

2. Let $\mathcal{E} = \{1, t, t^2\}$ be the standard basis for $\mathbb{P}_2$. Recall the matrix of T relative to $\mathcal{E}$ is given by

$$[T]_{\mathcal{E}} = [[T(1)]_{\mathcal{E}} \ [T(t)]_{\mathcal{E}} \ [T(t^2)]_{\mathcal{E}}]$$

. We compute the columns of the matrix: $T(1) = 0$, and hence $[T(1)]_{\mathcal{E}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$; $T(t) = 1$, and

hence $[T(t)]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$; and $T(t^2) = 2t$, and hence $[T(t^2)]_{\mathcal{E}} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$. Thus

$$[T]_{\mathcal{E}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

3. *Proof.* Let $P = [b_1 \ \dots \ b_n]$, then $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for $\mathbb{R}^n$ since P is invertible. We now compute the matrix of T relative to $\mathcal{B}$.

$$\begin{aligned} [T]_{\mathcal{B}} &= [[T(b_1)]_{\mathcal{B}} \ \dots \ [T(b_n)]_{\mathcal{B}}] \\ &= [[Ab_1]_{\mathcal{B}} \ \dots \ [Ab_n]_{\mathcal{B}}] \\ &= [P^{-1}Ab_1 \ \dots \ P^{-1}Ab_n] \\ &= P^{-1}A[b_1 \ \dots \ b_n] \\ &= P^{-1}AP \\ &= P^{-1}(PDP^{-1})P = D. \end{aligned}$$

Thus T has a diagonal representation if A is similar to a diagonal matrix D . □

4. Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x) = Ax$ and $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$.

- (a) Recall the matrix of T relative to basis $\mathcal{B}$ is given by $[T]_{\mathcal{B}} = [[T(b_1)]_{\mathcal{B}} \ [T(b_2)]_{\mathcal{B}}]$.

First note

$$T(b_1) = Ab_1 = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

$$\text{and hence } [T(b_1)]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}.$$

Second note

$$T(b_2) = Ab_2 = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix} = 3 \begin{bmatrix} -1 \\ 2 \end{bmatrix},$$

and hence $[T(b_2)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Thus the matrix of T relative to basis $\mathcal{B}$ is given by

$$[T]_{\mathcal{B}} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}.$$

(b) By the previous part, we know that $A = P_{\mathcal{B}}[T]_{\mathcal{B}}P_{\mathcal{B}}^{-1}$. Thus

$$\begin{aligned} A^{2019} &= A \cdot A \cdot \dots \cdot A && \text{2019 times} \\ &= (P_{\mathcal{B}}[T]_{\mathcal{B}}P_{\mathcal{B}}^{-1})(P_{\mathcal{B}}[T]_{\mathcal{B}}P_{\mathcal{B}}^{-1}) \cdots (P_{\mathcal{B}}[T]_{\mathcal{B}}P_{\mathcal{B}}^{-1}) && \text{2019 times} \\ &= P_{\mathcal{B}}[T]_{\mathcal{B}}[T]_{\mathcal{B}} \cdots [T]_{\mathcal{B}}P_{\mathcal{B}}^{-1} && \text{2019 times} \\ &= P_{\mathcal{B}}[T]_{\mathcal{B}}^{2019}P_{\mathcal{B}}^{-1} \\ &= \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 5^{2019} & 0 \\ 0 & 3^{2019} \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} -3^{2019} + 5^{2019} & \frac{-3^{2019}}{2} + 5^{2019} \\ 2 \cdot 3^{2019} - 5^{2019} & 3^{2019} - 5^{2019} \end{bmatrix}. \end{aligned}$$